

Electronic Limit Order Book and Order Submission Choice Around Macroeconomic News

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Abstract

We investigate the dynamics of the displayed part of the Island ECN limit order book for the equity index-linked securities market around macroeconomic news releases. Our results indicate that the quality of the electronic market, measured by spread and depth, deteriorates during the releases. Furthermore, they show that traders appear to switch to more aggressive strategies as early as three minutes before an announcement and are most aggressive within the first minute after the announcement. Also, there is a noticeable difference in the response of the limit order book to the macroeconomic news during pre-market hours relative to regular trading hours.

Keywords: macroeconomic announcements, limit order book, adverse selection

JEL classification: D81, E44, G10, G14

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1. Introduction

As electronic trading systems continue to gain market share worldwide, an understanding of their market structure becomes important. One of the controversial issues in this area is whether electronic markets can sustain a supply of liquidity in the absence of market makers with affirmative obligations. In the absence of designated intermediaries, limit orders are the only source of liquidity. But a limit order is subject to adverse selection risk due to its option-like properties. As a result, during periods of high uncertainty driven by the arrival of a large amount of new information, liquidity in electronic markets can dry up if traders switch from limit to market orders.

Before the mergers with Instinet Electronic Communication Network (ECN) and Nasdaq, Island Electronic Communication Network (Island ECN) was one of the largest electronic markets in the U.S. At their discretion, traders in the Island ECN provided all the liquidity in the electronic book through limit orders.

In this paper, we examine the effect of scheduled macroeconomic news releases on the dynamics of the displayed electronic limit order book by exploring the unique market structure of the Island ECN. To our knowledge, this is the first study that analyzes the dynamics of a purely electronic market before and after the macroeconomic announcements in the U.S. The detailed data on Island ECN enables us to analyze the dynamics of quoted spread, depth at and away from the best quotes, and the order submission behavior of the traders. Furthermore, the extended trading hours of the Island ECN allow us to study market reaction under different trading regimes.

We concentrate our analysis on the impact of macro news on an index-linked security, the Nasdaq-100 Index Trust Shares. This work contrasts with other studies that investigate the

state of the limit order book during public news releases in equity markets and focus primarily on the effect of firm-specific news on the market. For example, Ranaldo (2008) studies the reaction of the limit order book to firm-specific news on the Paris Bourse. Kavajecz (1999) investigates the state of the NYSE limit order book for the cross-section of stocks during earnings announcements and one type of macroeconomic announcement. The nature of adverse selection in the market for an index-linked “basket” security around macroeconomic announcements is different from that in the market for a single stock around firm-specific news. The major source of adverse selection during a firm-specific public announcement is the asymmetric information that arises due to the presence of private knowledge about the firm’s cash flows (see Green (2004)). In the market for a basket security, the losses incurred by the uninformed liquidity traders when trading with informed traders should be lower than in the market for an individual security (see Subrahmanyam (1991)). Further, during rapid, large price movements due to the arrival of new public information, limit order submitters risk being “picked-off” even in the absence of privately informed investors. Studies by Biais, Glosten, and Spatt (2005), Copeland and Galai (1983), Foucault, Roël, and Sandås (2003), and Fahlenbrach and Sandås (2003)) discuss this type of risk.

We note that our analysis relates to the displayed part of the Island’s limit order book. Certain data limitations make it difficult to draw any inferences about the hidden depth in the book. Yet we believe that the analysis of the displayed part of the limit order book is important because the displayed order book is what traders actually see when they make their order submission decisions.

The paper is organized as follows. In section 2 we formalize the hypotheses. Section 3 describes the data and the sample. In section 4 we report our results. Section 5 concludes.

2. Hypotheses

Theoretical and empirical studies of prescheduled anticipated public announcements indicate that market dynamics change both before and after the announcement (see, for example, Kim and Verrecchia (1991), Graham, Koski, and Loewenstein (2006)). In particular, macroeconomic news releases are accompanied by rise in price volatility (Ederington and Lee (1993), Erenburg, Kurov, and Lasser (2006)), fast price changes, and increase in spreads (Fleming and Remolona (1999), Frino and Hill (2001)).

To test the effect of macro news on limit order book in the market for Nasdaq-100 Index Trust Shares we formalize two hypotheses:

H1: The displayed spread increases around the time of macroeconomic news releases.

H2: The displayed depth decreases around the time of macroeconomic news releases.

The dynamics of limit order book, including changes in spreads and depth, are affected by traders' order submission behavior, which in turn is determined by price dynamics and the state of the limit order book. Current studies establish a connection between traders' order choice, price dynamics, and a state of the limit order book. For example, Bae, Jang, and Park (2003) find that traders in the NYSE place more limit orders when they expect high transitory volatility and when the spread is large. Biais, Hillion, and Spatt (1995) find that traders in the Paris Bourse submit more limit orders when order book is thin and bid-ask spread is large. Goldstein and Kavajecz (2004) document dramatic changes in limit order submission behavior of NYSE traders during extreme market movements.

The basic dilemma for traders is the choice between submitting limit orders rather than market orders. In reality, traders have a broader choice of submission strategies varying in degree

of aggressiveness with which traders pursue the execution of an order. For example, a limit order submitted with a limit price that improves on the current best bid or offer price is more aggressive than a limit order submitted away from the best bid-ask price, because it has much shorter waiting time and higher chances of being executed. According to Foucault's (1999) dynamic equilibrium model, higher volatility forces market participants to submit less aggressive orders. The spread and the transaction costs for market orders increase. Thus, traders prefer to submit limit orders. Therefore, higher volatility results in a higher proportion of limit orders.

The releases of macroeconomic news accompanied by fast changes in price and price volatility will affect traders' submission behavior. Hence, we state our third hypothesis:

H3: Traders follow less aggressive submission strategies around the time of macroeconomic news releases.

Studies that investigate the performance of an electronic limit order book often explore its side-by-side operation with officially designated human market makers. In some environments traditional markets with officially designated market makers are more resistant to informational shock. For example, Goldstein and Kavajecz (2004) show that during the extreme movements of the U.S. market in October 1997, the costs of providing liquidity through the electronic limit order book became so high that traders preferred to trade on the NYSE floor rather than through the SuperDot electronic limit order book.

In other environments, the existence of privileged traders such as market makers can increase the adverse selection costs in the market during the public news releases and discourage regular traders from trading right after news releases. Barclay, Hendershott, and McCormick (2003) show that Nasdaq traders are more likely to route orders to an ECN when there are high levels of information asymmetry, trading activity, and return volatility.

Island ESN operated under two trading regimes. During the regular market hours, the Island ECN competed for order flow with traditional market systems that employed designated human market makers. However, in the pre-market hours most trading occurred on the all-electronic markets and was considerably less active than during the regular market hours.¹ We can compare the reaction of Island's traders to the macroeconomic announcements over both time periods. Therefore, we formalize the fourth hypothesis:

H4: The reaction of the limit order book to the macroeconomic news shocks is most pronounced during the pre-market hours when the electronic market is the primary trading venue and the overall market is thin.

3. Sample and data

3.1. Island ECN

We obtain order data from the Island ECN for the period from January 1, 2002 to September 22, 2002. The data after September 22nd, 2002 was unavailable for us since, in response to a regulatory constraint, Island stopped displaying its limit order book for selected securities starting from September 23rd, 2002. The sample covers a total of 181 trading days.

The Island ECN was an electronic matching system that accepted, matched, and cancelled orders. In 2002, Island was the most actively traded ECN with considerable market share in Nasdaq traded securities. According to the Island's press release on May 15, 2002, the Island ECN had 11.2% of the Nasdaq share volume and 21.2% of the Nasdaq trade volume. Island also traded Amex and NYSE listed securities, as well as single stock futures. The ECN started operating in 1997 and was one of the first equity markets to provide a display of the limit

¹ Aside from Island, trading before the open also occurred on other electronic platforms, such as, for example, Instinet and Archipelago. In the pre-market hours some trading also occurred in the interdealer markets.

order book to all customers on an equal basis. In 2002, Island ECN completed its merger with the Instinet Group and has been renamed INET ATS (Alternative Trading System), Inc. On December 8, 2005, INET completed its merger with Nasdaq.

Trading on Island was available from 7:00 a.m. to 8:00 p.m. ET. Each participant could submit only limit orders, no market orders. However, a market order could be imitated by a limit order with a price that ensures its crossing with another order in the book. The top 15 quotes in the limit order book were publicly displayed on Island's website. In addition, traders had the option of submitting "non-displayed" limit orders that were not visible to the other participants and were not shown in the limit order book. However, non-displayed orders lost priority to displayed orders with the same price. Orders that were executed immediately were charged \$0.0019 per share, while orders that were added to the book received a rebate of \$0.0011 per share (the trading fees changed during the time period covered in our analysis) as a compensation for providing liquidity. Over 2002, the minimum price increment (tick size) for all the orders submitted to Island was \$0.001.

3.2. Procedure of constructing limit order book

Island's historical order data are available in text files formatted according to Island's proprietary ITCH data feed specification and are identical to the real-time data supplied to traders subscribing to Island. ITCH daily files comprise Add Orders, Modify Orders and Trade type messages that provide information about all submitted limit orders. An Add Order message describes a new order that has been accepted by the system. This message contains a time stamp, information about the buy-sell direction of the order, the number of shares added to the book, the stock symbol, and the unique reference number for this order. Modify Order messages include

Order Executed and Order Cancel messages and signal any changes in orders that were previously added to the book. An Order Executed message is issued if either a partial or a complete order is executed. It contains a time stamp, the reference number of the order that was executed (previously reported in the Add Order message), and the number of shares executed. Similarly, an Order Cancel message contains a time stamp, the reference number of the order and the number of shares being cancelled. An Add Order message is never sent for non-displayed orders. To indicate the execution of non-displayed orders, Island's trading system generates a Trade message. If a hidden order is executed in multiple parts, then the system issues several Trade messages. Each Trade message contains a time stamp, the reference number of the executed order, the buy-sell direction of the executed order, the incremental number of shares executed, the stock symbol, and the price.²

As noted above, the investor who submits the order can imitate a market order by submitting a limit order with a price that ensures its crossing with another order in the book. However, in the presence of hidden orders, the investor cannot always correctly determine what order submission price will make a limit order and what order price will make a market order. Thus, in our analysis we include only "obvious market orders," i.e., those orders that are identified by an Order Executed Message.

The ITCH database allows us to completely reconstruct Island's displayed limit order book at any point in time during the trading session. Using ITCH data files, we can track each displayed order and its modifications from the moment it is submitted through the time when it is either executed or cancelled.

² For full specification of the ITCH files refer to the Nasdaq website. Hasbrouck and Saar (2004) provide a summary of Island's background information and information about its trading protocol.

For every trading day, all outstanding limit orders on the Island ECN expire at 8:00 p.m. and are removed from the book. That simplifies the procedure of constructing a complete limit order book for any time during the day. For each day we start with an empty book and add all orders submitted from the start of the day to the time under consideration. After we remove all orders that have been executed or cancelled from the start of the day until the time under consideration, what remains are the displayed orders outstanding in the book at this specific time.

3.3. Nasdaq-100 Trust Shares

We focus our analysis on the market for the Nasdaq-100 Trust Shares. Unofficially, Nasdaq-100 Trust Shares are called “Cubes” due to their ticker symbol (QQQ).³ This equity basket security is similar to the shares of a regular index fund whose value follows the market index. However, just like a regular stock, Cubes are traded continuously throughout the day. These unique characteristics make the security extremely popular among investors: in August of 2002, the trading volume in Cubes was 3% of the combined trading volume for all equity securities traded on NYSE, Amex, and Nasdaq. This volume made the Cubes the most actively traded security in the U.S. equities market at that time.

In 2002, the Island ECN carried a significant portion of the QQQ trading volume. In the first three weeks of September, Island’s market share in the QQQ was 36.9%.⁴ Figure 1 shows the intraday minute-by-minute Cubes’ volume and the number of trades made on Island for the year 2002. Volume spikes occur at 8:30 a.m., 10:00 a.m., and 12:00 p.m. when most macroeconomic news is released. Generally, it takes about ten minutes for the volume to return to its pre-announcement level.

³ Currently, Cubes are listed on Nasdaq with a corresponding ticker symbol QQQQ. At the time of our analysis, the Cubes were listed on Amex and had corresponding three-letter ticker symbol QQQ. We use the original notations.

⁴ American Stock Exchange (<http://amex.com>).

[Figure 1 about here]

Figure 1 demonstrates the differences between pre-market hours and regular trading hours. Trading activity in the QQQs increases during regular trading hours (9:30 a.m. to 4:00 p.m.) when all the major traditional stock exchanges are open. However, when all the major traditional stock exchanges are open, Island ECN carries only a portion of all trading in this stock.

3.4. Macroeconomic announcements

The time of the news releases varies across types of macroeconomic announcements, but it is the same for a specific type of announcement. We obtain the dates and times of news releases from the database on macroeconomic announcements and forecasts collected by Money Market Services (MMS) International, a subsidiary of Informa Financial Group. All of the announcements in our sample are issued on a monthly basis. Since the most important news is released at either 8:30 a.m. or 10:00 a.m., we concentrate our analysis on market dynamics around these times.

4. Results

4.1. Macroeconomic news and price volatility

Table 1 shows lists of announcements included in our study, the corresponding time of release, and the number of releases over the sample period. To estimate the impact of each type of announcement on the market price, we follow Fleming and Remolona's (1997) method. We use the following regression:

$$K_{it} = \gamma_0 + \sum_{t=1}^{T-1} \gamma_t D_t + \sum_{m=1}^M \eta_m D_{mit} + e_{it}, \quad (1)$$

where K_{it} is price volatility measured as absolute return on day i in the one-minute interval t ; D_{mit} is a dummy variable that is equal to one if the announcement m is made on day i in the beginning of interval t ; and D_t is a dummy variable used to control for the intraday variation in volatility. We compute volatility as absolute value of the log price change times 100. We compute price as the mid-quote based on the best displayed quotes. Table 1 reports coefficients η_m and the corresponding statistics. Five announcements have the largest impact on the prices: Consumer Confidence, Employment, Durable Goods Orders, Real GDP, and Retail Sales. The average impact of each of these announcements is statistically significant and ranges from 22.9 to 38.6 basis points. The average impact of most other announcements is not significant and does not exceed 10 basis points. Overall, the results reported in Table 1 are consistent with the results of Ederington and Lee (1993) and Fleming and Remolona (1997) for interest rates and foreign exchange markets and of Erenburg et al. (2006) for the U.S. equity markets.

[Insert Table 1 about here]

4.2. Quoted spreads around 8:30 a.m. and 10:00 a.m. announcements

In our analysis of the displayed spreads and depth, we compare the state of the limit order book during 8:30 a.m. and 10:00 a.m. on the days when the corresponding announcements were made to the days without any announcements. Over our sample period of 181 trading days there are 90 days when at least one announcement has been made at 8:30 a.m. and 52 days when at least one announcement has been made at 10:00 a.m.

Figure 2 presents second-by-second quoted spreads during 8:30 a.m. and 10:00 a.m. for announcement and non-announcement days. Spreads widen within one minute before the 10:00

a.m. announcements and within three minutes before the 8:30 a.m. announcements, and then decrease to pre-announcement levels within three to four minutes after the announcements. Spreads are about three cents at 8:30 a.m. on non-announcement days and rise to almost six cents during the announcement on announcement days. Spreads at 10:00 a.m. are on average 1.6 cents when there are no announcements and reach 2.5 cents during the announcement time on the days when the news is released.

[Figure 2 about here]

Table 2 presents average time-weighted quoted one-minute spreads. Announcement day spreads are significantly larger than the spreads on non-announcement days from one minute before to three and four minutes after the 8:30 a.m. and 10:00 a.m. respectively. The results on spreads support our first hypothesis and are in line with the previous studies of the impact of macroeconomic news on various markets.

[Table 2 about here]

In contrast to human-intermediated markets with designated market makers such as the dealer markets investigated by Fleming and Remolona (1999) or the open outcry futures market analyzed by Frino and Hill (2001), electronic markets have a somewhat different mechanism for forming spreads. The spreads are not set by designated market makers, but are the result of collective and offsetting actions by traders (see, for example, Foucault's (1999) model). In such markets, bid-ask spreads can widen for several reasons. First, because traders can submit their limit orders further away from the mid-quote without changing the proportion of submitted limit orders to market orders. Second, traders can submit more market orders instead of limit orders. Third, traders can start cancelling more limit orders. Fourth, specifically for the Island ECN

market, a larger portion of limit orders can be hidden, which exacerbates the effect of the increasing displayed spreads.

4.3. Book's depth

In addition to the quoted spreads, another important liquidity dimension is the level of depth at and away from the inside quotes. Kavajecz (1999) shows in his study that both NYSE specialists and limit order traders reduce depths during four announcements made by the Federal Open Market Committee concerning the changes in the Fed Funds rate. However, the NYSE is a market system that operates differently from a pure electronic limit order book. The presence of specialists who might be more informed than other traders during times of high information uncertainty can adversely affect the traders' willingness to provide liquidity at these times.

To evaluate the changes in displayed depth, we follow Goldstein and Kavajecz (2004). We measure the cumulative displayed depth in the limit order book at three levels: at the quotes (best bid and best offer within the book), within one cent away from the quotes, and in the rest of the limit order book. This measure of depth controls for changes in the inside spread in contrast to another commonly used measure, the cumulative depth relative to the mid-quote. For example, when the spreads widen, our measure of depth will be higher than the measure of depth relative to the midquote.

Figure 3 shows the cumulative displayed depth around 8:30 a.m. and 10:00 a.m. on announcement days. Depth decreases considerably within four minutes before 10:00 a.m. announcements and rises to the original levels within ten minutes after the announcements. The changes in depth level around 8:30 a.m. announcements are not as dramatic, even though the

most important announcements in terms of their impact on equity market prices are made at 8:30 a.m.

[Figure 3 about here]

Table 3 provides the displayed cumulative depth for each minute within a 20-minute interval around the announcements. The most significant changes in depth during the 8:30 a.m. and 10:00 a.m. announcements occur at the mid level, for orders submitted within one cent from the quotes. Also the depth significantly decreases for orders with prices more than one cent away from the quotes during the 10:00 a.m. announcements. We do not observe any significant decreases in the quoted depth. In fact, the quoted depth slightly increases during 10:00 a.m. announcements.

[Table 3 about here]

The results generally support our second hypothesis for both times of news releases, although the evidence is substantially weaker for 8:30 a.m. announcements.

Overall, the dynamics of the displayed spread and depth suggest that the market quality of the visible part of the order book during the macroeconomic announcements deteriorates. These results are in accordance with Lee, Mucklow, and Ready (1993), who argue that an increase in spreads and a decline in depth together imply deterioration in liquidity.

4.4. Order aggressiveness

Here, we investigate how traders change their order submission strategies in response to macroeconomic announcements. In our analysis, we follow a classification of order submission strategy by aggressiveness similar to the one developed by Biais, Hillion, and Spatt (1995).

Table 4 provides summary statistics of five different types of orders submitted within 20 minutes

around 8:30 a.m. and 10:00 a.m. Our analysis comprises only the orders that traders decide to display to the public. BBO refers to the prevailing quotes, the bid, and the offer prices of the most competitive displayed limit orders.

The most aggressive orders are market orders. They are generally smaller than limit orders and make up less than 10% of the total number of orders submitted around 8:30 a.m. and 10:00 a.m. The average size of a limit order is much larger, ranging from 2224.83 to 4870.11 shares per order. Limit orders submitted within and at the prevailing displayed quotes are considerably smaller than the limit orders submitted outside of the quotes. Most orders are submitted outside of the prevailing quotes; 44.64% around 8:30 a.m. and 54.64 % around 10:00 a.m. Following Biais et al. (1995) and Rinaldo (2004), we treat order cancellations as the least aggressive type of order. Table 4 shows that after the limit orders submitted outside of prevailing quotes, the cancellations are the second largest group of orders by their average size and number of submissions. In addition, we find that the orders submitted around 10:00 a.m. are larger than the orders submitted around 8:30 a.m. for any type of order.

[Table 4 about here]

For further analysis, we partition the market orders into two size groups relative to the median order size. We classify orders that are submitted around 8:30 a.m. as large orders if they are larger than 200 shares. We classify market orders that are submitted around 10:00 a.m. as large market orders if they are larger than 500 shares. The results are not qualitatively sensitive to the classification method.

To test whether order aggressiveness changes during macroeconomic news releases, we adopt the ordered probit regression method introduced by Rinaldo (2004). Equations 2 and 3 describe our model:

$$z_t^* = \sum_{i=1}^k \alpha_i x_{i,t} + \sum_{l=-10}^{10(l \neq 0)} \lambda_l d_{l,t} + \varepsilon_t, \quad (2)$$

$$z_{n,t} = \begin{cases} 1 & \text{if } -\infty < z_t^* \leq \mu_1, \\ m & \text{if } \mu_{m-1} < z_t^* \leq \mu_m, \text{ for } m = 2, 3, 4, 5, \\ 6 & \text{if } \mu_5 < z_t^* \leq \infty. \end{cases} \quad (3)$$

In these equations, $z_{n,t}$ is a dependent variable that takes discrete values according to a degree of aggressiveness of specific order type. $z_{n,t}$ is equal to one for the most aggressive large market order, $z_{n,t}$ is equal to two for the less aggressive smaller market order, $z_{n,t}$ is equal to three for limit orders with prices better than displayed best quotes, $z_{n,t}$ is equal to four for limit orders with prices matching the best quotes on the same side, $z_{n,t}$ is equal to five for limit orders with prices away from the best quotes on the same side, and $z_{n,t}$ is equal to six for the cancellations. $d_{l,t}$ is a dummy variable equal to one if the order has been submitted within the l -th minute of the announcement time on the scheduled day of announcement; $x_{i,t}$, for $i=1,2,3$, refers to control variables; ε_t is distributed normally across observations; μ_m refers to the probit thresholds. The control variables “Wait”, “Spread”, and “Volatility” are similar to the ones employed by Ranaldo (2004). The larger or smaller number of lagged returns used to compute measure of transitory volatility does not change the results qualitatively. We do not include the quoted depth as a control variable because, according to the previous subsection results, the quoted depth does not change significantly during the announcements.

Table 5 reports the estimation results of the ordered probit model described by equations 2 and 3. It is important to note that negative coefficients indicate positive relations between the order aggressiveness and the independent variable. The coefficient estimates for the dummy variables corresponding to two minutes and less before 8:30 a.m. and one minute and less before

10:00 a.m. become negative, significant, and greater in the absolute value than coefficients for earlier time intervals. This result indicates that traders start acting more aggressively in anticipation of the news. The spike in order aggressiveness happens in the first minute after the announcement. The effect from the announcement lasts for another four to five minutes. The comparison of the estimates for the full model with the estimates for the model without the control variables⁵ confirms the robustness of the results to the endogenous changes in the limit order book. Overall, the results reject Hypothesis 3, indicating that during both 8:30 a.m. and 10:00 a.m. announcements, traders become considerably more aggressive in their order submission behavior.

[Table 5 about here]

Table 5 documents the increase in order aggressiveness, but it does not show us what drives the changes in order aggressiveness. Does order aggressiveness increase because traders switch from limit orders to market orders, because they submit more limit orders at or better than the outstanding quotes, or because traders cancel more limit orders? We estimate how the probability of submitting each type of order changes during the announcement by computing the marginal effects of the announcement dummy variables. Using the estimates of the probit model described by equations 2 and 3, the cumulative probabilities can be estimated as follows:

$$\begin{aligned}
 \Pr[z_{it} = 1] &= \Phi(\hat{\mu}_1 - \sum_{i=1}^k \hat{\alpha}_i x_{i,t} - \sum_{l=-10}^{10(l \neq 0)} \hat{\lambda}_l d_{l,t}), \\
 \Pr[z_{it} = m] &= \Phi(\hat{\mu}_m - \sum_{i=1}^k \hat{\alpha}_i x_{i,t} - \sum_{l=-10}^{10(l \neq 0)} \hat{\lambda}_l d_{l,t}) - \Phi(\hat{\mu}_{m-1} - \sum_{i=1}^k \hat{\alpha}_i x_{i,t} - \sum_{l=-10}^{10(l \neq 0)} \hat{\lambda}_l d_{l,t}) \quad \text{for } m = 2, 3, 4, 5, \\
 \Pr[z_{it} = 6] &= 1 - \Phi(\hat{\mu}_5 - \sum_{i=1}^k \hat{\alpha}_i x_{i,t} - \sum_{l=-10}^{10(l \neq 0)} \hat{\lambda}_l d_{l,t}),
 \end{aligned} \tag{4}$$

⁵ Results from the model without the control variables are available on request.

$\Phi(\cdot)$ is the cumulative normal distribution function. The marginal effect for each dummy variable corresponding to a specific time interval is estimated by comparing the probabilities that result when the dummy takes its two different values while the other independent variables are held at their sample means. (See Ranaldo, 2004.) By construction, the sum of the marginal effects across all order types for any time interval (dummy) must be zero.

Table 6 reports the marginal effects of announcements on the probability of each type of order in each time interval from ten minutes before to ten minutes after the announcement.⁶ The results indicate that during both 8:30 a.m. and 10:00 a.m., changes in trader order preferences are qualitatively similar. In particular, traders are less likely to cancel orders on announcement days during news releases. For example, within the first minute after the 8:30 a.m. announcement the probability that a trader will cancel an order is 12.77% lower than on a non-announcement day over the same time interval and it is 10.86% lower in the first minute after the 10:00 a.m. announcement. The results also show that during announcements, traders submit relatively more market and limit orders. The largest increase in probabilities occurs for large market orders and limit orders submitted within the prevailing displayed quotes.⁷

[Table 6 about here]

Figure 4 shows the relative changes in the submission rates for five types of orders submitted around 8:30 a.m. announcements. (The results for 10:00 a.m. announcement are very similar). The relative changes in the submission rates for large market orders are the largest, followed by small market orders, and then by limit orders submitted within and at the prevailing

⁶ For computing the marginal effects in Table 6 we use estimates from the probit model without the control variables. Results from using the estimates of the full model are similar.

⁷ Table 6 shows changes in probabilities, but not the rates of change in probabilities. For example, the fact that the probability of submitting a limit order within the prevailing quotes in the first minute after the 8:30 a.m. announcement increases by a larger percentage than the probability of submitting a large market order (4.26% versus corresponding 3.45%) does not mean that the traders switch their preferences from large market orders to aggressive limit orders.

quotes. The submission rates of limit orders with the price outside of the prevailing quotes stay almost unchanged. Around the announcements, traders cancel fewer orders and increase order submission aggressiveness by submitting more market orders and aggressively priced limit orders.

Because we lack data on non-displayed order submissions, the results of our analysis in this section must be interpreted with caution. The results refer only to the part of order submission behavior that traders are willing to display to the public.

4.5. Pre-market hours vs. regular trading hours

The results in our previous sections indicate a substantial difference in the reaction of the limit order book to the news released at 8:30 a.m. relative to 10:00 a.m. Our last hypothesis, regarding the differences in book reaction to the news under different trading sessions, is supported by the results for the spreads in Table 2, but not by the results for depth in Table 3. The spreads increase more dramatically during macroeconomic announcements made during the pre-market hours, but the depth dynamics is more pronounced during the news released during market hours. However, we note that the types of announcements released during 8:30 a.m. and 10:00 a.m. are not the same. Therefore, some of the differences in the book reaction to the 8:30 a.m. and 10:00 a.m. news might be due to the difference in types of announcements and not to the difference in trading sessions.

It is difficult to compare the changes in traders' aggressiveness between two trading sessions. Traders switch their order preferences in a qualitatively similar fashion. As noted in the previous section, around 8:30 a.m. and 10:00 a.m., traders switch to more aggressive types of orders. However, the results of Table 5 suggest that the minute-by-minute dynamics of order

aggressiveness are not identical across two different trading sessions. Table 7 presents the statistics for testing the differences in limit order book reaction to the macroeconomic announcements released during two different trading sessions. Overall, hypothesis 4 is rejected for depth and order aggressiveness and is supported for spreads.

[Table 7 about here]

4.6. *Hidden orders*

Electronic markets allow participants to submit “iceberg,” or partially displayed limit orders. For these orders, the traders see price and only a portion of the order volume. Island ECN allowed the subscribers to submit “non-displayed” orders that were completely hidden from traders and other market participants. The available data only includes information about the executions of non-displayed orders from the book and does not contain any information about hidden orders that were never executed. The data also do not identify what type of incoming order triggered the transaction. As a result, we lack information on hidden order submissions and cannot observe the complete order submission choice. However, we can provide an analysis of the executed non-marketable orders. As opposed to the marketable limit orders designed to replicate market orders and priced to execute immediately, non-marketable orders entered the book before being executed. A trade message from ITCH files identifies what type of non-marketable order has been executed – displayed or non-displayed. The following statistics are reported for the non-marketable limit orders.

For the Nasdaq 100 Tracking Stock, the average proportion of executed non-marketable hidden orders in all executions is 50% and 70% around 8:30 a.m. and 10:00 a.m., respectively. Although we believe that this study provides an important contribution to understanding the

performance of electronic markets, it is important to discuss the role of hidden orders in Island ECN.

A major advantage of the hidden order is that it eliminates the risk of exposure in the presence of high information asymmetry. In the presence of high information asymmetry, traders prefer to submit more of their limit orders in the non-displayed format to avoid the premature exposure of their intent. However, non-displayed orders have lower priority than regular displayed orders and have to be priced more aggressively. As a result, traders have to weigh the advantage of non-displaying their order with the necessity to price these orders more aggressively during announcements. Furthermore, the risk that traders face during macroeconomic announcements is mostly related to the quick change in the market valuation, and this type of risk is similar for the submitters of both hidden limit orders and regular visible limit orders.

For single stocks, according to Hasbrouck and Saar (2004), executed hidden orders account for about 12% of all order executions and executed shares. During regular trading hours in the market for the Nasdaq 100 Index Tracking Stock (QQQ) executed hidden orders make up to 80% of the volume (number of shares), and up to 70% of the number of all trades.

The high proportion of hidden orders in the market for the Nasdaq 100 Index Tracking Stock seems to contradict the properties of the basket security described by Subrahmanyam (1991). He argues that most of the trading in basket securities is generated by liquidity traders rather than by informed traders. Consequently, if the only use of a hidden order is to decrease the premature exposure of an informed order, it is not clear why the proportion of hidden orders in the Nasdaq-100 Index Tracking Stock is so much higher than the proportion of hidden orders in the market for individual stocks.

It seems that the amount of trading activity is at least as important in explaining the usage of hidden orders as the proportion of informed trading. For example, the hidden order activity is also high for other types of actively traded market index linked securities such as S&P Depositary Receipts (SPY) tracking S&P 500 market index or Diamonds Trust Series 1 (DIA) security that tracks DJIA index. Furthermore, in the market for Cubes, the proportion of executed hidden orders is extremely high only during regular market hours when trading activity is also high. Before 8:00 a.m. and after 4:15 p.m. executed limit orders make only 10-20% of the volume.

The Island's rebate structure provides an incentive for the traders to submit hidden orders. As described previously, incoming orders that are executed immediately are charged 0.19 cents per share, but orders added to the book receive a rebate of 0.11 cents per share at execution as a compensation for providing liquidity. Traders who want fast execution but prefer to avoid charges for immediate execution try to submit hidden orders very close to the opposite side of the book, so that the order will be executed once the new aggressive order is submitted from the opposite side. This way, the submitter of a hidden order obtains fairly fast execution and earns the rebate, while a submitter of a visible order bears the risk of accidentally hitting the hidden order from the other side and paying a fee.⁸

Based on the discussion above, it does not appear that the unavailability of the data on hidden orders can bias the results of our study in any particular direction. But, it is clear that hidden orders on Island ECN play a very large role in the market for actively traded equity-index linked securities.

4.7. Data limitations

⁸ We thank anonymous referee for pointing out this explanation.

One may argue that the dataset used in the study has several potential drawbacks. We do not believe that these data limitations undermine the validity of our results or conclusions. However, we discuss these limitations below and leave it up to reader to make a final judgment. One concern may be related to the short sample period. Our sample period covers less than nine months, which limits the number of announcements to less than ten for each type of macroeconomic announcement. However, the reader should note that our analysis includes 16 different types of announcements (six made at 10:00 a.m. and ten made at 8:30 a.m.). Except for the preliminary analysis, where we estimate the individual effect of each type of announcement on price volatility, in the rest of the paper we test the average effect of any macroeconomic news released at 8:30 a.m. and at 10:00 a.m. Thus, we have a sufficiently large number of days when at least one announcement has been made and when no major announcement has been made.⁹

Another concern may be that the market experienced a sharp downfall during the sample period, and hence our results may not be generalized for the expanding or neutral markets. However, we do not analyze the direction of the impact of market news on the market. Instead, we make a comparison between the market dynamics on the days when the news is released and the days when no news is released.

The lack of data on non-displayed orders submissions limits the scope of our analysis. However, as discussed in the previous section, it does not appear that this would bias our conclusions in any particular direction. Traders in the electronic market infer all the information from the displayed part of the limit order book. Therefore, by investigating the dynamics of the displayed part of the limit order book, we gain insight into the traders' decision-making process.

⁹ See section 4.2.

5. Conclusion

We investigate the dynamics of the Island ECN electronic limit order book during the releases of macroeconomic news in the market for the Nasdaq 100 equity-index tracking stock. The results of the analysis demonstrate deterioration in limit order book quality as measured by spreads and full depth in response to a macroeconomic announcement.

The reaction of the limit order book to the announcements differs across trading sessions. During the pre-market hours, the increase in displayed spreads captures most of the reaction to the macroeconomic news. But during regular market hours, we find a decrease in visible depth and an increase in the limit order book spreads in reaction to a news release.

The results also suggest an increase in the aggressiveness of displayed orders submission during the macroeconomic news releases. The major change in traders' behavior is observed in the first minutes after the announcement. We observe the peak in order aggressiveness in the first minute after the announcement. The effect from announcement lasts for another four to five minutes. Overall, our results indicate that around macroeconomic news releases, traders prefer to cancel fewer orders and submit relatively more market and aggressively priced limit orders. The changes in order submission dynamics around macroeconomic announcements are similar during different trading periods.

Our results show that while the quality of the limit order book deteriorates during the macroeconomic news shocks, electronic markets are not only a viable competitor to the traditional human-mediated markets, but can stand on their own in the absence of traditional markets.

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Table 1

Average impact of macroeconomic announcements on the prices of Nasdaq 100 index-tracking stock (QQQ) for the period from January 1, 2002, to September 20, 2002.

$$K_{it} = \gamma_0 + \sum_{t=1}^{T-1} \gamma_t D_t + \sum_{m=1}^M \eta_m D_{mit} + e_{it},$$

Rank	Announcement	Release time (ET)	Num. of announc.	Coefficient
1	Consumer Confidence	10:00	8	0.386***
2	Nonfarm Payrolls (Employment)	8:30	9	0.298***
3	Durable Goods Orders	8:30	8	0.293***
4	Real GDP	8:30	8	0.287***
5	Retail Sales	8:30	9	0.229**
6	Index of Leading Indicators	10:00	9	0.099*
7	Factory Orders	10:00	9	0.067
8	NAPM	10:00	9	0.056
9	Producer Price Index	8:30	9	0.054
10	Housing Starts	8:30	9	0.048*
11	Personal Consumption Expenditures	8:30	8	0.043
12	Construction Spending	10:00	9	0.005
13	Consumer Price Index	8:30	9	-0.011
14	New Home Sales	10:00	8	-0.012
15	Business Inventories	8:30	9	-0.016
16	Goods & Services Trade Balance	8:30	9	-0.019*
Adjusted R ²				0.0179

K_{it} is price volatility measured as absolute return on day i in the one-minute interval t ; D_{mit} is a dummy variable that is equal to one if the announcement m is made on day i in the beginning of interval t ; and D_t is a dummy variable used to control for the intraday variation in volatility. The coefficients represent the average impact of the announcement on the return volatility for the QQQ's prices. We compute volatility as absolute value of the log price change times 100. We compute price as the mid-quote based on the best displayed quotes. t -statistics are based on the Newey and West (1987) heteroskedasticity and autocorrelation consistent covariance matrix.

*, **, *** indicate that the impact on volatility is statistically significant at 10%, 5%, and 1% levels, respectively, in a two-tailed test.

Table 2

Time-weighted mean quoted spreads over one-minute intervals for announcement and non-announcement days for the period from January 1, 2002 to September 20, 2002.

Event time interval, min	Around 8:30 a.m.		Around 10:00 a.m.	
	Announc. Day	Non-announc. Day	Announc. Day	Non-announc. Day
(-10; -9)	3.46	3.35	1.65	1.56
(-9; -8)	3.41	3.30	1.64	1.59
(-8; -7)	3.54	3.29	1.58	1.56
(-7; -6)	3.66*	3.26	1.58	1.55
(-6; -5)	3.37	3.21	1.60	1.55
(-5; -4)	3.36	3.21	1.62	1.57
(-4; -3)	3.47**	3.07	1.56	1.56
(-3; -2)	3.56	3.24	1.66	1.55
(-2; -1)	3.67	3.28	1.59	1.54
(-1; 0)	4.28***	3.30	1.74*	1.55
(0; 1)	5.00***	3.48	2.26***	1.70
(1; 2)	3.93**	3.25	2.04***	1.62
(2; 3)	3.43***	3.06	1.79**	1.60
(3; 4)	3.31	3.06	1.74*	1.60
(4; 5)	3.18	2.96	1.65	1.61
(5; 6)	3.01	3.06	1.65	1.57
(6; 7)	2.90	3.01	1.68	1.57
(7; 8)	2.82	3.11	1.64	1.58
(8; 9)	2.79	3.00	1.67	1.55
(9; 10)	2.92	2.92	1.62	1.57

We use the z-statistics from the median test of the difference between two independent samples to compare trading activity on announcement and non-announcement days.

*, **, *** - indicate 0.05, 0.01 and 0.001 significance levels respectively.

Table 3

Displayed cumulative depth (1000 shares) over one-minute intervals for the three levels of limit order prices: at the best bid and offer (BBO); at the prices up to one cent away from the best displayed BBO (BBO; ± 0.01); in the rest of the limit order book, at prices more than one cent away from the best bid and offer (± 0.01 ; $\pm \infty$).

Panel A. Displayed Depth Around 8:30 a.m. Announcements						
Event time interval, min	BBO		(BBO; ± 0.01)		(± 0.01 ; $\pm \infty$)	
	Ann.	No Ann.	Ann.	No Ann.	Ann.	No Ann.
(-10; -9)	5.152	4.939	7.211	5.640	70.442	70.535
(-9; -8)	4.900	5.023	5.596	5.096	71.274	70.692
(-8; -7)	5.194	4.422	5.613	5.203	71.297	70.645
(-7; -6)	4.946	4.599	5.945	4.926	71.030	71.375
(-6; -5)	5.443	4.501	5.530	4.940	71.692	71.668
(-5; -4)	5.659	5.043	5.663	4.844	71.360	72.748
(-4; -3)	5.153	4.808	5.643	5.307	71.966	74.240
(-3; -2)	4.876	5.064	4.770*	5.483	72.077	73.737
(-2; -1)	4.730	4.298	4.005**	5.720	72.528	74.382
(-1; 0)	4.523	4.757	2.987***	5.711	73.482	74.343
(0; 1)	4.457	4.884	3.740***	5.489	73.759	75.201
(1; 2)	5.019	4.755	4.420*	5.617	79.182	76.879
(2; 3)	4.684	4.999	5.280	6.360	81.612	77.412
(3; 4)	4.929	5.588	5.921	6.782	83.440*	76.898
(4; 5)	5.304	5.226	6.694	6.661	83.982	79.849
(5; 6)	5.184	5.080	6.266	6.714	84.568	81.864
(6; 7)	4.749	5.417	6.621	6.729	83.961	83.859
(7; 8)	5.676	5.413	7.653	7.155	85.962	84.091
(8; 9)	5.162	4.665	6.658	7.212	85.508	85.722
(9; 10)	5.120	4.764	7.003	7.272	86.136	89.670
Panel B. Displayed Depth Around 10:00 a.m. Announcements						
(-10; -9)	7.415	7.805	24.637	24.242	322.406	319.990
(-9; -8)	8.754	8.205	25.190	24.312	333.109	322.356
(-8; -7)	8.044	8.061	25.132	24.816	336.319	329.288
(-7; -6)	9.019	8.510	25.943	25.602	343.573	333.366
(-6; -5)	8.592	8.392	25.092	25.270	343.554	338.000
(-5; -4)	8.282	8.269	24.608	26.013	340.444	342.908
(-4; -3)	8.185	8.337	23.443	26.021	340.495	344.420
(-3; -2)	8.582	8.098	22.701	25.315	333.923	342.584
(-2; -1)	9.071*	8.124	18.398**	25.391	332.237**	342.565
(-1; 0)	8.404	8.677	14.131***	23.918	292.365***	341.273
(0; 1)	7.142	8.660	9.927***	24.067	236.895***	325.536
(1; 2)	8.428	8.646	14.042***	24.191	270.418***	339.245
(2; 3)	7.495*	8.700	16.562***	24.427	296.405**	348.563
(3; 4)	8.204	8.388	18.839**	25.646	313.249*	353.321
(4; 5)	8.870	8.363	22.464	25.945	329.571	358.229
(5; 6)	8.305	8.486	23.464	25.554	334.590	360.762
(6; 7)	8.008	8.357	22.975	26.670	337.640	362.190
(7; 8)	8.618	8.117	24.903	26.018	338.822**	365.422
(8; 9)	9.147	8.238	25.061	25.499	346.832*	368.428
(9; 10)	9.876*	8.608	24.380	25.671	352.459	368.587

We average the depth separately across days on which there was at least one type of macroeconomic news released (Ann.) and for the days when no announcements were scheduled (No Ann.). We use the median test of the difference between two independent samples to compare displayed depth on announcement and non-announcement days.

*, **, *** - indicate 0.05, 0.01 and 0.001 significance levels respectively.

Table 4

Summary statistics of five different types of orders submitted within 20 minutes during 8:30 a.m. and 10:00 a.m. BBO refers to the prevailing best displayed quotes.

	Around 8:30 a.m.					Around 10:00 a.m.				
	Mean size	Median size	St. Dev.	Total Number	Percent, %	Mean size	Median size	St. Dev.	Total Number	Percent, %
Market orders	612.70	200.00	1049.00	17563	5.79	791.34	500.00	1326.00	262502	8.8
Limit orders within BBO	2641.05	1600.00	3082.72	25840	8.52	3196.30	1100.00	4543.24	180980	6.07
Limit orders at BBO	2224.83	1000.00	3335.14	11513	3.8	2681.54	1000.00	4292.00	109700	3.68
Limit orders outside of BBO	3172.75	2000.00	2917.43	135412	44.64	4664.61	3200.00	6206.30	1629864	54.64
Cancel	3064.26	2000.00	2850.53	113014	37.26	4870.11	2400.00	7242.84	800123	26.82

Table 5

Estimation results for the ordered probit model.

$$z_t^* = \sum_{i=1}^k \alpha_i x_{i,t} + \sum_{l=-10}^{10(l \neq 0)} \lambda_l d_{l,t} + \varepsilon_t,$$

$$z_{n,t} = \begin{cases} 1 & \text{if } -\infty < z_t^* \leq \mu_1, \\ m & \text{if } \mu_{m-1} < z_t^* \leq \mu_m, \text{ for } m = 2, 3, 4, 5, \\ 6 & \text{if } \mu_5 < z_t^* \leq \infty. \end{cases}$$

Indep. Variables	8:30 a.m. announcements	10:00 a.m. announcements
λ_{-10}	-0.073 *	0.028 ***
λ_{-9}	-0.052 **	0.024 ***
λ_{-8}	0.006	0.037 ***
λ_{-7}	0.021	0.046 ***
λ_{-6}	0.019	0.006
λ_{-5}	0.017	0.039 ***
λ_{-4}	-0.028 *	0.006
λ_{-3}	-0.034 **	0.022 ***
λ_{-2}	-0.094 ***	-0.009
λ_{-1}	-0.088 ***	-0.111 ***
λ_{+1}	-0.363 ***	-0.378 ***
λ_{+2}	-0.182 ***	-0.243 ***
λ_{+3}	-0.109 ***	-0.109 ***
λ_{+4}	-0.067 ***	-0.072 ***
λ_{+5}	-0.037 ***	-0.063 ***
λ_{+6}	-0.013	-0.034 ***
λ_{+7}	-0.011	-0.037 ***
λ_{+8}	0.005	-0.008
λ_{+9}	-0.005	-0.028 ***
λ_{+10}	0.032 **	-0.025 ***

The dependent variable takes discrete values according to a degree of aggressiveness of specific order type; λ_l is a coefficient for dummy variable that is equal to one if the order has been submitted within the l -th minute of the announcement time on the scheduled day of announcement, and zero otherwise. To save space, we do not report the estimates of threshold values μ and independent control variables. These estimates are available on request.

*, **, *** - indicate 0.05, 0.01 and 0.001 significance levels respectively.

Table 6

Estimates of marginal changes in probabilities during the announcements for five types of orders.

$$\begin{aligned}\Pr[z_{it} = 1] &= \Phi(\hat{\mu}_1 - \sum_{i=1}^k \hat{\alpha}_i x_{i,t} - \sum_{l=-10}^{10(l \neq 0)} \hat{\lambda}_l d_{l,t}), \\ \Pr[z_{it} = m] &= \Phi(\hat{\mu}_m - \sum_{i=1}^k \hat{\alpha}_i x_{i,t} - \sum_{l=-10}^{10(l \neq 0)} \hat{\lambda}_l d_{l,t}) - \Phi(\hat{\mu}_{m-1} - \sum_{i=1}^k \hat{\alpha}_i x_{i,t} - \sum_{l=-10}^{10(l \neq 0)} \hat{\lambda}_l d_{l,t}) \quad \text{for } m = 2, 3, 4, 5, \\ \Pr[z_{it} = 6] &= 1 - \Phi(\hat{\mu}_5 - \sum_{i=1}^k \hat{\alpha}_i x_{i,t} - \sum_{l=-10}^{10(l \neq 0)} \hat{\lambda}_l d_{l,t}),\end{aligned}$$

Event time interval, min	8:30 a.m. Announcements						10:00 a.m. Announcements					
	Large MO	Small MO	LO within BBO	LO at BBO	LO outside of BBO	Cancel	Large MO	Small MO	LO within BBO	LO at BBO	LO outside of BBO	Cancel
(-10; -9)	0.55	0.33	0.83	0.28	0.75	-2.74	-0.26	-0.17	-0.20	-0.10	-0.19	0.92
(-9; -8)	0.38	0.23	0.58	0.19	0.55	-1.93	-0.22	-0.15	-0.17	-0.09	-0.16	0.80
(-8; -7)	-0.04	-0.03	-0.06	-0.02	-0.07	0.22	-0.34	-0.23	-0.27	-0.13	-0.26	1.24
(-7; -6)	-0.14	-0.09	-0.23	-0.08	-0.25	0.79	-0.41	-0.29	-0.33	-0.16	-0.33	1.52
(-6; -5)	-0.13	-0.08	-0.21	-0.07	-0.22	0.71	-0.06	-0.04	-0.04	-0.02	-0.04	0.20
(-5; -4)	-0.12	-0.07	-0.19	-0.06	-0.20	0.65	-0.35	-0.24	-0.28	-0.14	-0.28	1.30
(-4; -3)	0.20	0.12	0.31	0.10	0.31	-1.05	-0.05	-0.04	-0.04	-0.02	-0.04	0.18
(-3; -2)	0.25	0.15	0.38	0.13	0.37	-1.28	-0.20	-0.14	-0.16	-0.08	-0.15	0.73
(-2; -1)	0.72	0.43	1.06	0.35	0.93	-3.49	0.08	0.06	0.07	0.03	0.05	-0.29
(-1; 0)	0.67	0.40	1.00	0.33	0.88	-3.28	1.14	0.74	0.82	0.39	0.42	-3.51
(0; 1)	3.45	1.85	4.26	1.31	1.90	-12.77	4.81	2.73	2.82	1.27	-0.77	-10.86
(1; 2)	1.49	0.86	2.09	0.68	1.54	-6.66	2.79	1.70	1.82	0.85	0.22	-7.37
(2; 3)	0.84	0.50	1.23	0.41	1.06	-4.04	1.12	0.72	0.81	0.39	0.42	-3.45
(3; 4)	0.50	0.30	0.76	0.25	0.70	-2.52	0.72	0.47	0.53	0.26	0.34	-2.32
(4; 5)	0.27	0.16	0.41	0.14	0.40	-1.37	0.63	0.41	0.46	0.23	0.31	-2.03
(5; 6)	0.10	0.06	0.15	0.05	0.15	-0.50	0.33	0.22	0.25	0.12	0.18	-1.09
(6; 7)	0.08	0.05	0.12	0.04	0.12	-0.40	0.35	0.24	0.27	0.13	0.20	-1.19
(7; 8)	-0.03	-0.02	-0.05	-0.02	-0.05	0.17	0.07	0.05	0.06	0.03	0.05	-0.26
(8; 9)	0.03	0.02	0.05	0.02	0.05	-0.17	0.27	0.18	0.21	0.10	0.16	-0.91
(9; 10)	-0.22	-0.13	-0.35	-0.12	-0.38	1.20	0.24	0.16	0.18	0.09	0.14	-0.82

We estimate the marginal effect for each dummy variable corresponding to a specific time interval by comparing the probabilities that result when the dummy takes its two different values while we hold the other independent variables at their sample means.

Table 7

Comparing the reaction of limit order book to the announcements under different trading sessions: pre-market hours compared to regular market hours.

Event time interval, min	Spreads (Chi-square test)	Depth (Chi-square test)	Aggressiveness (Z-test)
(-10; -9)	0.07	0.46	2.51*
(-9; -8)	1.13	0.13	3.29***
(-8; -7)	2.76	0.36	1.53
(-7; -6)	0.26	0.05	1.29
(-6; -5)	0.19	0.04	-0.67
(-5; -4)	3.09	0.08	1.18
(-4; -3)	0.58	0.23	1.90
(-3; -2)	1.72	0.56	3.28***
(-2; -1)	7.78**	9.96**	5.05***
(-1; 0)	11.04***	32.71***	-1.28
(0; 1)	0.98	20.58***	-1.38
(1; 2)	0.98	15.19***	-5.69***
(2; 3)	0.31	10.29**	0.01
(3; 4)	0.92	4.64*	-0.41
(4; 5)	0.67	3.38	-2.22*
(5; 6)	1.81	3.29	-1.64
(6; 7)	4.19*	3.42	-2.00*
(7; 8)	3.53	1.66	-0.95
(8; 9)	0.07	0.73	-1.73
(9; 10)	1.47	0.74	-4.02***

Statistics for spreads and depth (chi-square) test for the difference in spread and depth reactions between the two trading sessions. Statistics for aggressiveness (z-test for λ_l coefficients from Eq. (2) and (3)) offer a test for difference in aggressiveness changes under two different trading sessions. We compute depth as sum of all displayed limit orders within each one minute interval.

*, **, *** - indicate 0.05, 0.01 and 0.001 significance levels respectively.

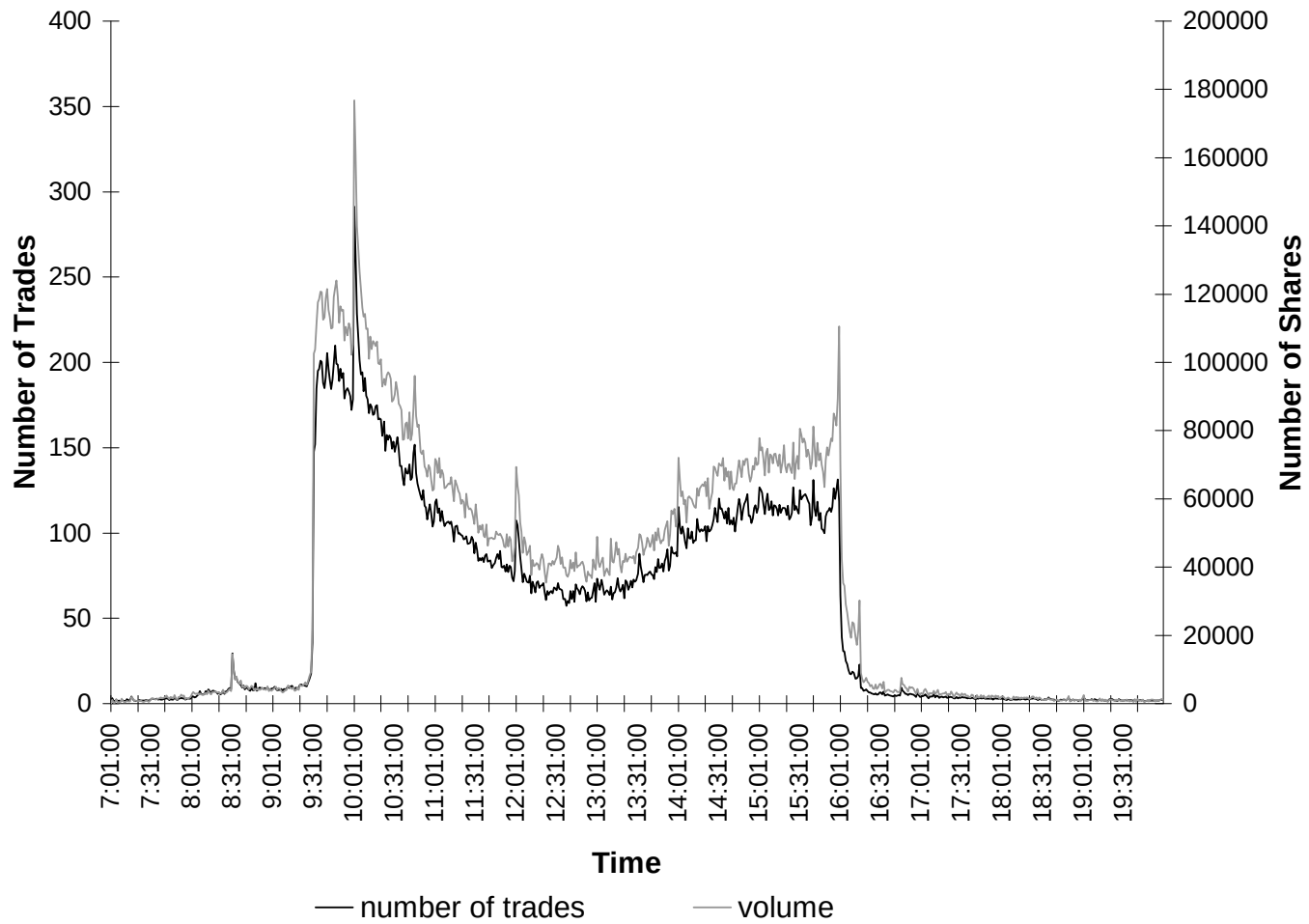


Fig. 1. Number of trades and volume for QQQ over the whole year 2002.

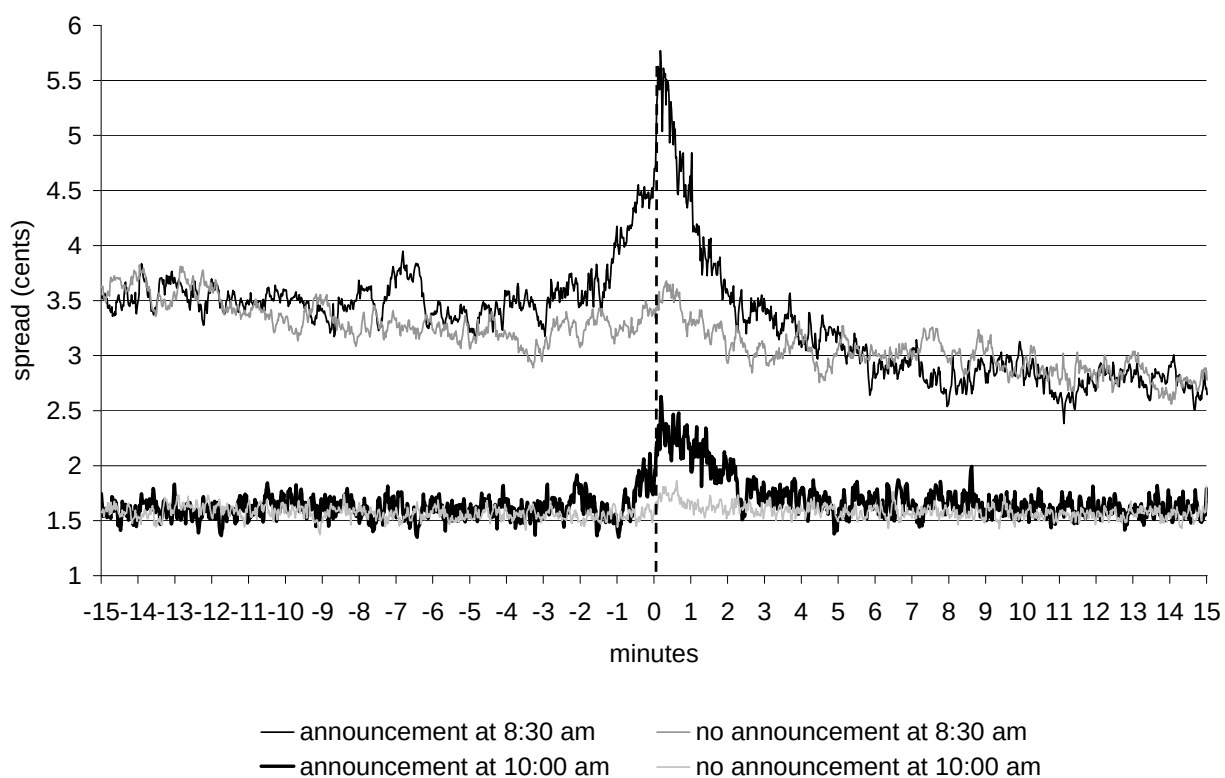


Fig. 2. Quoted displayed spreads for QQQ.

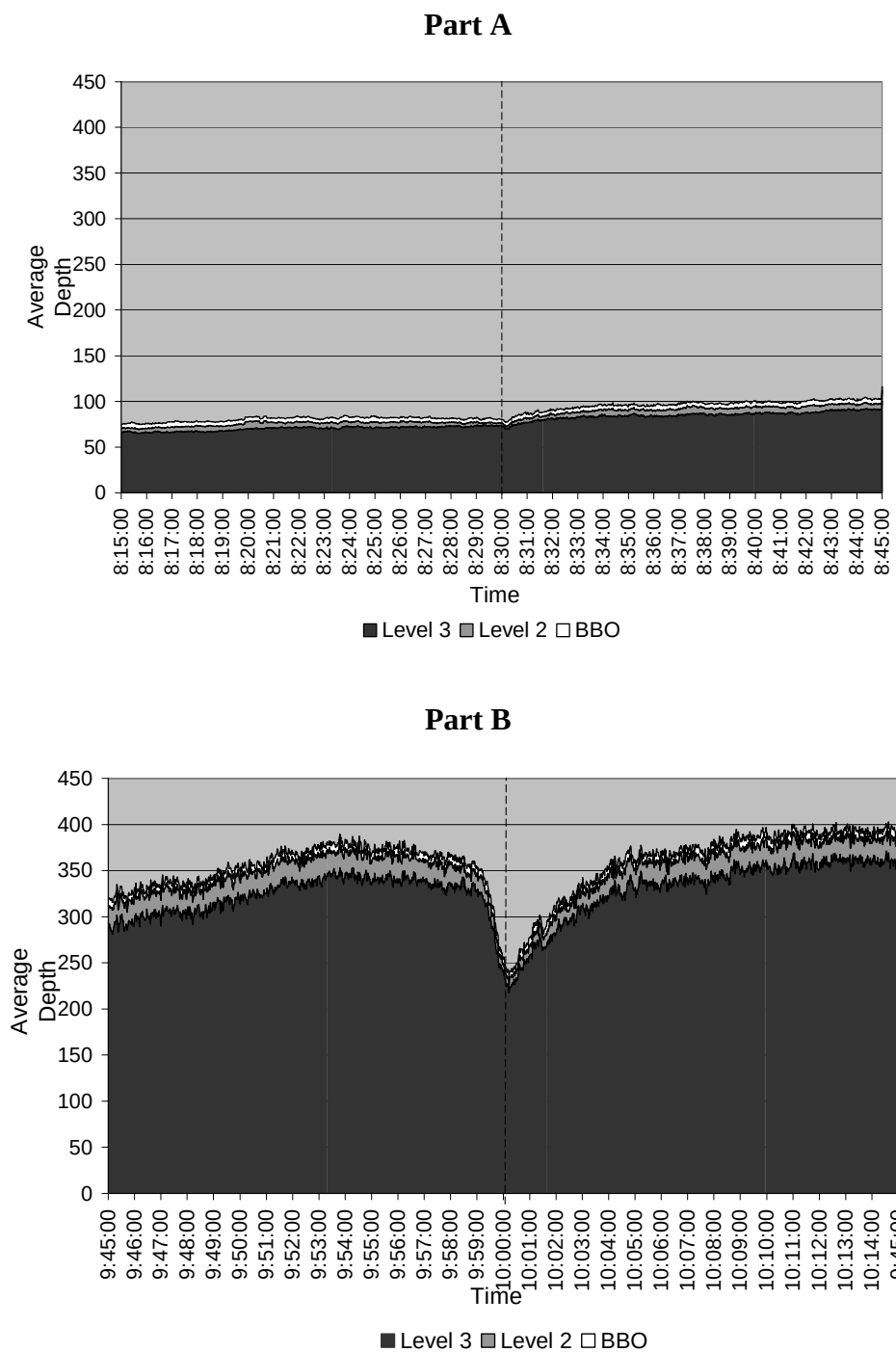


Fig. 3. Cumulative displayed depth on announcement days. Part A: around 8:30 a.m. Part B. Around 10:00 a.m.

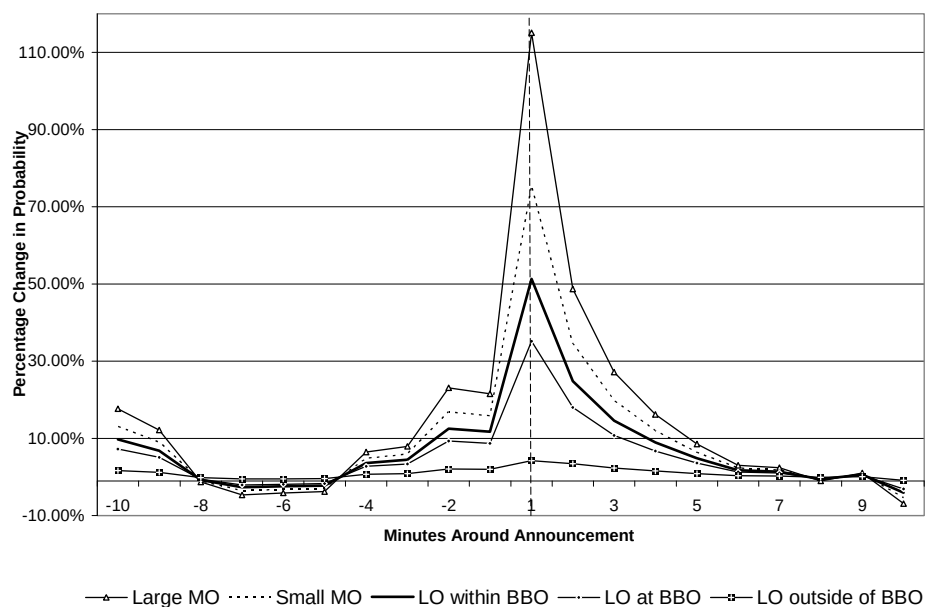


Fig. 4. Relative changes in the submission rates around 8:30 a.m.